The Wallet Without a Polity: Displaced and Stateless Subjects under a Bonded Settlement Primitive

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Abstract

The bonded settlement primitive of the mechanism paper, implemented and verified in the implementation paper, exhibits a property whose consequences are most consequential not for populations conventionally treated by labor or commercial law, but for those who occupy the negative space of those frames: refugees, stateless persons, undocumented migrants, and populations whose residency is beyond the reach of stable banking and enforcement infrastructure. The primitive's precondition is a cryptographic key, not a recognized civil-legal subjecthood. This paper develops the consequence.

The wallet-as-legal-subject argument extends here in a different direction. In the labor setting the wallet collapses the Roman *res*/*persona* distinction so that the productive asset is simultaneously property and party; in the present setting it allows the productive asset to act as party where civil-legal subjecthood is itself absent. We do not claim that the primitive resolves statelessness. Arendt [1951] named the right to have rights as the fundamental human right — a right whose denial is a civilizational failure rather than an individual misfortune — and the technological infrastructure considered here cannot grant it. What it grants is narrower: a capacity to have commerce, where the capacity to have rights is denied. The two should not be confused. For populations for whom the first is denied, the second is non-trivial.

The argument is positional rather than programmatic. Real constraints — key custody, counterparty willingness, fiat on-ramp, collateral access — are named and not dismissed. The analytical claim is that the primitive moves the humanitarian-economics gap from an architectural impossibility to an operational engineering question, and that this is a consequential move worth scholarship.

Keywords: statelessness, refugee economies, displaced populations, the wallet as legal subject, Arendt, humanitarian economics, commerce-without-jurisdiction

1 Introduction

The legal infrastructure of commerce presumes a recognized civil subject. Contract law requires parties whose standing the forum will accept; banking access requires identity documents the domestic regime issues; commercial-dispute adjudication requires a court whose jurisdiction reaches both sides. Each of these presumptions is so deeply embedded in the architecture of contemporary trade that its failure mode — populations for whom the presumptions do not hold — is treated by the architecture as exception rather than as a system property.

The United Nations Refugee Agency estimated 120 million forcibly displaced persons worldwide at the end of 2023, and approximately 4.4 million persons registered as stateless, with a

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substantially larger estimated population whose statelessness is undocumented [UNHCR, 2024]. An overlapping and considerably larger population lacks stable banking and enforcement access for reasons of informal migration status, residence without papers, or residency in jurisdictions whose financial infrastructure does not reach them. Across these categories, the productive capacity of well over a hundred million working-age adults is systematically underused because the legal scaffold for engaging it is absent. The humanitarian-economics literature on refugee economies [Betts and Collier, 2017] and the policy frontier between camp and labor market [Pritchett, 2006] documents the cost of this gap and the existing repertoire of responses.

This paper argues that the bonded settlement primitive of the mechanism paper, implemented and verified in the implementation paper, changes the architectural posture of the gap. The primitive operates on parties whose existence as parties does not require civil-legal recognition. A wallet can sign; a bonded commitment can settle; an exchange can complete — without a host-state court, a domestic bank, or a permission-to-work visa sitting between the parties. The labor-law consequences of the primitive are addressed elsewhere; the present paper develops the consequence for populations to whom the labor-law debate is a second-order question, because the first-order question is whether they are recognized parties to any contract in any forum at all.

What this paper is not. This paper does not claim that the primitive resolves statelessness. It does not confer citizenship, restore legal recognition, or create enforcement against a state that denies a person's standing. It is not a substitute for the political work that addresses those gaps — work that remains necessary, that this paper does not duplicate, and that no technological infrastructure can perform. It does not predict adoption rates, endorse any particular humanitarian program, or assess the political consequences of widespread wallet adoption among displaced populations. Its argument is structural: the architectural posture of the commerce-access gap changes once a settlement primitive whose precondition is a cryptographic key exists.

Paper organization. Section 2 summarizes the settlement primitive in the terms relevant here. Section 3 restates the wallet-as-legal-subject argument from the labor-law paper in the form needed for the present paper. Section 5 develops the central claim across four moves: scale of the affected population, the architecture they face, what the primitive does and does not do, and the Arendtian distinction between the right to have rights and the capacity to have commerce. Section 6 names the operational constraints the primitive does not dissolve. Section 7 concludes.

2 The Settlement Primitive

We summarize only the features relevant to the present argument. The mechanism is established in the mechanism paper; the Solidity implementation and formal verification in the implementation paper.

The primitive composes two mechanisms. *Asymmetric bonding* makes defection more expensive than cooperation at every bonded edge. *Buyer dominance with atomic resolution* settles every active order in a process simultaneously, at the buyer's signature, with no third-party adjudicator. The first mechanism removes the trusted enforcer; the second removes the trusted clearing house. For the present paper, both are load-bearing: the displaced or stateless subject lacks access to either, and the primitive's contribution is to make both unnecessary.

A transaction between two parties is consummated by dual-signed EIP-712 commitments. Each party signs from their own wallet. Each party locks collateral against their own performance — the buyer locks $2P$, the seller locks $2G$ where G is the cumulative upstream value bonded in the process. Only the buyer can trigger resolution. Defection is structurally more expensive than cooperation. There is no platform between the parties, no arbitrator authorized to resolve disputes on-chain, and no admin key permitting external override.

The features that matter for the present paper are three. First, the kernel does not require either party to be a recognized civil subject; it requires them to control a key that can sign. Second, the kernel does not require a forum to enforce the agreement; the collateral structure is the enforcement mechanism. Third, the kernel does not route value through a banking intermediary; settlement is direct, wallet to wallet, atomic on resolution.

3 The Wallet as Legal Subject (Brief)

Roman private law distinguished *res* (things, property, objects that can be owned and alienated) from *persona* (persons, parties, subjects capable of acting legally). The distinction survived and structures modern private law. The bonded primitive collapses the *persona*/*res* partition: a wallet is simultaneously a property (a key-pair owned and transferable) and a party (a signer, bonder, settlement-receiver). The collapse is operational rather than metaphorical; it follows from the architecture of dual-signed bonded commitments.

For the present paper the relevant consequence is narrower than the labor-law one but harder. In the labor setting the *persona*/*res* collapse dissolves the subordination gradient between employer and employee, because each party signs for itself and bonds for its own performance. In the present setting — where one or both parties may lack stable *persona* status in the host jurisdiction at all — the collapse means something stronger: the wallet provides a party-to-the-transaction whose existence does not depend on the prior existence of a recognized civil subject. The labor-law argument shows that the wallet can be both property and party. The argument here is that the wallet can be a party where no civil-legal party exists.

This is not a claim that the wallet *is* a civil-legal subject. Civil-legal subjecthood is conferred by polities, and its absence is a political condition. The claim is narrower: a party to a bonded commitment can transact, settle, and accumulate verifiable performance history without first acquiring civil-legal subjecthood, because the kernel does not consult the polity for permission to settle a bond.

4 Self-Sovereignty: The Cryptographic Key as Precondition

The previous section established that the wallet can be a party where no civil-legal party exists. The deeper observation, which the self-sovereign-identity literature has been developing independently of the bonded-primitive context, is that this arrangement is a particular kind of agency that deserves a name distinct from both *civil-legal personhood* and the older *natural personhood* the philosophical tradition has analyzed under that heading. The name we adopt, following Allen [2016] and the broader self-sovereign-identity literature [Tobin and Reed, 2017, W3C, 2022], is *self-sovereignty*: the condition of an agent whose capacity to bind, transact, and accumulate verifiable history derives from cryptographic self-control rather than from recognition by a polity.

Three acts, three preconditions discharged. A bonded transaction reduces to three acts at the participant's side, and each act has historically required a civil-legal precondition that the bonded primitive replaces with a cryptographic one.

Signing. The act of binding oneself to terms has historically required a recognized capacity to enter contracts — adulthood, sound mind, jurisdictional standing, and the authority to act for oneself in the relevant forum. Under the bonded primitive the capacity to sign is the capacity to control a private key. EIP-712 typed-structured-data signatures verify through public-key cryptography against the address that signed; the verification does not consult any registry of recognized subjects. A cryptographic key is the precondition; a passport is not.

Bonding. The act of pledging assurance has historically required a counterparty willing to extend credit, a court willing to enforce a guarantee, or a recognized residency status enabling a banking relationship. Under the bonded primitive the act of pledging assurance is the act of locking collateral in a permissionless token. The token's transferability is enforced by the chain's consensus, not by any host institution. The bond's return on cooperation is enforced by the kernel, not by a court. A wallet that holds the bond-denomination currency can pledge without intermediation.

Settling. The act of receiving consideration has historically required a banking relationship with an institution that can credit the receiving party's account in the receiving jurisdiction's currency. Under the bonded primitive settlement is direct wallet-to-wallet on resolution. The chain is the clearing layer at the architectural level. The architectural property is unconditional; the operational property in any given denomination is not, since the bond-denomination currency is itself a jurisdictional artifact (USDC is freezable by Circle on Treasury request; ETH is acquired through KYC-gated exchanges). The bond-currency embedding constraint is developed in Section 6 below; here we register that the architectural elimination of the banking precondition is real but its practical realization for any specific participant remains conditional on the bond-currency layer's own jurisdictional reach.

The three acts together constitute a complete commercial participation. The three preconditions historically required for the acts — recognized capacity, intermediated assurance, banked settlement — are each replaced by cryptographic substitutes that do not require recognition by the civil-legal order. This is what self-sovereignty in our usage means: the agent's commercial agency is constituted by what the agent can prove cryptographically rather than by what the agent has been recognized as politically.

What self-sovereignty does and does not mean. The term has been used in several adjacent senses, and we owe the reader a precise scope.

The *self-sovereign-identity* (SSI) literature [Allen, 2016, Tobin and Reed, 2017] is grounded in identity specifically: the participant holds verifiable credentials about themselves rather than depending on identity providers to attest to their identity. Allen [2016] articulates ten principles for self-sovereign identity — existence, control, access, transparency, persistence, portability, interoperability, consent, minimization, and protection — several of which extend beyond credential-holding into the agency-of-interaction register: control over how identity data is used, persistence across institutional change, portability across providers. The "operational self-sovereignty" sense we develop in this paper is the commerce-specific narrowing of Allen's broader agency-of-interaction principles, applied to the bonding, signing, and settling acts of bilateral commerce rather than to identity per se. The W3C Decentralized Identifiers specification [W3C, 2022] formalizes the architectural shape: identifiers controlled by the subject through cryptographic keys, resolvable to documents the subject publishes, with no central registrar. SSI in the W3C-DID register is about *identity*, and the bonded primitive does not implement identity (the kernel knows wallets, not persons; identity is a runtime-tier concern composed by parties who want it).

The *libertarian / Lockean* sense of sovereignty [Nozick, 1974, Locke, 1689] is normative: the agent has natural rights of self-ownership that political arrangements must respect or violate. The bonded primitive is silent on natural rights; it provides architecture, not ethics. A reader who takes self-sovereignty in this sense should read our usage as compatible with but not requiring the normative framework: the architecture supplies a substrate on which a self-ownership reading *can* be expressed, but the substrate makes no claim about whether self-ownership is morally fundamental.

The sense we adopt in this paper is narrower than either: an agent whose commercial agency is operationally constituted by cryptographic self-control rather than by polity recognition. This is a structural fact about how the bonded primitive operates, not a normative claim about how it should. The displaced participant who sign-bonds-settles a process is exercising self-sovereignty in this operational sense, regardless of how their natural-rights status is read by any external philosophical framework. The architectural fact is what we claim; the normative reading is an interpretation that readers in different traditions will compose differently.

Why this matters specifically for the displaced. The general operational sense of self-sovereignty applies universally: any wallet that signs-bonds-settles is exercising this kind of agency. But the marginal value of the architecture is asymmetric. A participant whose civil-legal subjecthood is already conferred and stable has plenty of paths through which to express their commercial agency; the bonded primitive is one substrate among many, and self-sovereignty in the operational sense adds modestly to a position that was already operational. A participant whose civil-legal subjecthood is denied or unstable does not have those paths. For them, the operational self-sovereignty the bonded primitive supplies *is* the agency, not an augmentation to it. The displaced or stateless participant whose signing-bonding-settling is enabled by nothing more than control of a private key has acquired a commercial-participation capacity that the surrounding infrastructure does not grant.

This is what the present paper has been calling the capacity to have commerce. Self-sovereignty is the precise architectural property that makes the capacity available to a population for which it was not previously available. The Arendtian distinction of the previous section — right to have rights versus capacity to have commerce — decomposes once self-sovereignty is named explicitly. The right to have rights is conferred by a polity recognizing the agent. Self-sovereignty is constituted by the agent's cryptographic control of their own commercial agency. The two are different things, and conflating them has been a recurring error in the literature this paper has been pushing back against in both directions: those who claim cryptographic infrastructure restores Arendtian rights overstate; those who claim cryptographic infrastructure cannot do meaningful work for the displaced because it does not restore Arendtian rights also overstate. Self-sovereignty is real, modest, and operationally load-bearing where conventional infrastructure fails; it is not a substitute for political belonging where conventional infrastructure succeeds.

Key custody as the load-bearing operational variable. Self-sovereignty's architectural cleanness should not obscure its operational fragility. The architecture turns on the participant's ability to hold and use a private key reliably. For the displaced and stateless populations of greatest interest to this paper, key custody is exactly where the architecture meets the hardest constraints: unstable access to power and connectivity, devices vulnerable to seizure or loss, environments where social-recovery networks are themselves disrupted by the displacement. Self-sovereignty in the operational sense collapses to the participant's actual capacity to retain exclusive control of their key over time. The Section 6 treatment of these constraints is therefore not parenthetical; it is where the architectural property cashes out into deliverable agency.

The self-sovereign-identity community has invested significant effort in mitigations: social-recovery wallets where participant-trusted contacts can collectively recover a lost key [Buterin, 2021, Argent, 2020, Safe, 2022], secure elements in inexpensive hardware, mnemonic backup standards encoded in widely-spoken languages, and emerging biometric-binding schemes that tie key control to a physical attribute. None of these is perfect; each is an engineering artifact with its own threat model. We do not endorse any specific mitigation here. We name the constraint because self-sovereignty without durable key custody is structurally incomplete, and the engineering of durable key custody for displaced populations is the work that translates the architectural availability into deliverable agency.

5 The Capacity to Have Commerce

5.1 Scale

Beyond the UNHCR figures cited in Section 1, the population relevant to the present argument is broader than the forcibly-displaced and stateless categories alone. Undocumented migrants who hold no recognized status in their country of residence, internally displaced persons whose host sub-jurisdiction does not extend banking infrastructure to their location, and populations in failed-state or contested-sovereignty zones whose nominal citizenship is not exercisable for commercial purposes all share the operationally-relevant property: they cannot reliably enter the legal infrastructure of commerce as recognized parties. Estimates of the combined population vary substantially with definition; the order of magnitude is hundreds of millions, not millions.

5.2 The Received Architecture

Labor law, contract law, banking, and commercial-dispute adjudication each presume a recognized legal subject attached to a jurisdiction that enforces their rights. The relevant populations occupy the negative space of that presumption.

- The employment relation is typically unavailable to them formally: permission-to-work requirements gate access to the employee category, and undocumented work, where it occurs, occurs outside legal recognition.
- The independent-contractor relation is unstable: cannot register a business in many jurisdictions of residence, cannot enforce contracts in local courts that do not recognize the party's standing, cannot reliably serve process across jurisdictional boundaries.
- Banking access is restricted or absent: opening an account typically requires documents of residency and identity that the population does not have or does not have in the form the host jurisdiction recognizes.
- Adjudication has no stable forum: commercial disputes have no local court willing to hear them from a party whose standing is contested.

The humanitarian-economics tradition has documented the cost of this gap. Betts and Collier [2017] argue that displaced populations are systematically prevented from contributing to the economies that host them by an institutional architecture designed for citizens, with the result that the host economy captures little of the productive capacity the displaced bring, and the displaced themselves remain dependent on humanitarian transfers when productive participation would be both possible and preferred. Pritchett [2006] makes the broader argument about labor-mobility friction and the welfare cost of the institutional gap between labor markets in origin and destination economies.

5.3 What the Primitive Does and Does Not Do

The primitive does not resolve statelessness. It does not confer citizenship, restore legal recognition, or create enforcement against a state that denies the person's standing. It is not a substitute for the political work that addresses those gaps. What it does is structurally narrower:

(a) It provides a commerce architecture whose precondition is a cryptographic key rather than a recognized civil-legal subjecthood. A wallet does not require a passport to bond; a bonded commitment does not require a host-state court to enforce; a settled exchange does not require a domestic bank to clear.

(b) It produces a verifiable performance record that travels with the wallet across jurisdictional boundaries. A contributor whose civil-legal subjecthood is contested in jurisdiction A accumulates the same on-chain attestation history as one whose status is settled in jurisdiction B. The record is portable in a way that conventional employment or contracting records are not.

(c) It opens a region of productive activity — bilateral, bonded, settled directly between parties — to participants for whom the legal scaffold that conventionally underwrites such activity is unavailable. The set of activities thus opened is bounded by what can be coordinated bilaterally and settled atomically; the bound is non-trivial but not all of productive activity.

These are descriptive properties of the architecture. They do not depend on any humanitarian program or any particular legal reform; they follow from the construction of the primitive.

5.4 Arendt and the Right to Have Rights

Arendt [1951] observed that the fundamental human right is the right to have rights — the right to belong to a political community within which one's other rights have standing. Her analysis of statelessness named the condition as a civilizational failure rather than an individual misfortune. The person without a polity is not merely deprived of specific rights; they are deprived of the institutional setting in which having-rights is even a coherent claim. The remedy is political, not technological.

The Arendtian objection to the present paper's reading is sharper than the surface formulation, and the paper is improved by engaging it directly. Part II Chapter 9 of *The Origins of Totalitarianism* ("The Decline of the Nation-State and the End of the Rights of Man") is not only a diagnosis of statelessness; it is an analysis of a specific kind of *worldlessness* — the loss not of particular entitlements but of the standing of the human person within a public world that confers significance on their actions. Arendt [1958]'s later distinction between *labor*, *work*, and *action* sharpens this: the world is constituted through action visible to others, in a public space within which the actor is recognized as an actor; loss of polity-membership is loss of the public space within which action is constitutive of the world.

The Arendtian objection to a "capacity to have commerce" framing is therefore not that commerce is unimportant for the displaced; it is that commerce-without-polity may be the deepening of worldlessness rather than its alleviation. To exchange goods with strangers across a substrate that confers no membership and recognizes no public standing is, on a strict Arendtian reading, to participate in *labor* (the metabolic necessity of staying alive) without re-entering *action* (the recognized public act). The bonded substrate looks, on this reading, like a more efficient form of bare life rather than a restoration of the world.

We take the objection seriously and decline it on these terms. The capacity-to-have-commerce we describe is not a substitute for membership; it is a structural precondition that membership historically subsumed and that, where membership is denied, can be supplied separately. The bonded transaction does include a recognition function in Arendt's sense, on a small scale: the commitment is dual-signed, the performance is attested, the record is public, and the counterparty acknowledges the participant as a party with whom binding agreements can be made. This is not the political recognition Arendt regarded as fundamental — the recognition of co-membership in a polity — but it is not nothing in her register either. It is bilateral recognition of capacity-to-bind, conferred not by a political authority but by a counterparty who has chosen to bond with the participant. The accumulated record of such recognitions across many bonded relationships does not constitute a polity, but it does constitute a public visibility of the participant's actions that statelessness conventionally denies. The paper's claim is therefore narrower than the Arendtian framing of restoration: the substrate supplies a portable evidentiary subjecthood that is not coextensive with political membership, but that fills a gap statelessness has historically left empty.

The bonded primitive does not grant the right to have rights; it grants, more narrowly, a capacity to have commerce. The two should not be confused. The right to have rights is political and is not within the gift of any technology. The capacity to have commerce is structural and becomes operational once the cryptographic infrastructure is in reach. For populations for whom the first right is denied, the second capacity is non-trivial; for populations for whom the first right is granted, the second capacity is largely subsumed by existing institutions and the primitive is one infrastructure among several. The distinction matters for the audience of this paper because the primitive's leverage is asymmetric: where the conventional infrastructure works, it adds modestly; where the conventional infrastructure fails, it adds the architectural condition for participation at all.

The protection-gap residue the substrate cannot reach. The 1951 Refugee Convention and its 1967 Protocol enumerate a set of rights that the host state owes to refugees within its territory: *non-refoulement* (Article 33), access to courts (Article 16), identity papers (Article 27), travel documents (Article 28), employment, public education, and public relief [Refugee Convention, 1951, Goodwin-Gill and McAdam, 2021]. The *Right to Have Rights* as Arendt names it is closer to the spirit of the Convention than to any single article; the Convention operationalizes the spirit through specific institutional duties [Kesby, 2012, Benhabib, 2004]. The bonded primitive supplies none of these: it cannot prevent *refoulement* (which is a sovereign decision on territorial admission); it cannot grant access to courts (which is a jurisdictional decision); it cannot issue identity papers, travel documents, or work permits. The protection gap that Goodwin-Gill and McAdam describe is real, vast, and political; the substrate addresses one of its many components (the inability to bind in commerce without a recognized counterparty) and leaves the others where they were.

A reader sympathetic to refugee policy should take this honestly: the present paper is not an argument for de-prioritizing the political work of Convention enforcement, host-state obligations, or international protection regimes. The substrate is a *useful adjunct* where it works, not a substitute for the broader regime. The position we hope to occupy is closer to Benhabib: the rights-of-others framework identifies layers of moral and legal claim that are differently secured by different institutional arrangements, and the bonded substrate adds one such layer (commerce-binding capacity) without displacing the others.

6 Operational Constraints

The present paper is positional rather than programmatic. The primitive moves the commerce-access gap from architectural impossibility to operational engineering question; it does not solve the engineering question. We name four constraints that remain.

Key custody. Populations with unreliable access to power, devices, and connectivity face usability barriers at the cryptographic layer that the primitive does not solve. Loss-of-key risk is high in unstable environments, and key-recovery infrastructure for displaced populations is not built. The constraint is real and evolves with the broader infrastructure around the protocol; solutions exist (social recovery, custodial wallets with revocable trust) but each carries trade-offs the population is poorly positioned to evaluate.

Counterparty willingness. A well-resourced buyer in a recognized jurisdiction may decline to bond with a counterparty whose civil-legal standing is contested. The decline is not because of the primitive; it is because of downstream legal exposure in the buyer's home jurisdiction (sanctions screening, anti-money-laundering obligations, forum-of-enforcement concerns). The primitive does not dissolve this friction. Its effect, where it has one, is to shift the friction from architectural — "commerce with the displaced is impossible" — to discretionary — "some counterparties will, others will not."

Fiat on-ramp. Converting wallet-denominated value to local currency still runs through existing rails, which remain gated for the population in question. The primitive operates within the crypto-native settlement layer; the bridge between that layer and local purchasing power remains a separate problem with its own architecture and its own gatekeepers. Stablecoin acceptance by local merchants, where it occurs, partially closes this gap; full closure depends on infrastructure outside the primitive.

Collateral access. Bonding requires collateral. A counterparty without starting capital cannot bond. This is a hard constraint on the primitive's applicability to populations whose displacement is correlated with asset loss. Possible mitigations exist at the assembly tier — sponsor pools, cooperative bonding, assembly-aggregated bonds, microfinance-style group guarantees — but each of these is a construction on top of the primitive, not a property of the primitive itself, and none are yet built at scale. This constraint is not minor: the primitive's equilibrium results assume both parties are legally and materially free to enter the bonding game — that entering is a choice, and that capital sufficient to post the required bond is available. For displaced populations whose displacement is itself the loss of capital, the assumption fails on the second clause. The primitive is symmetric in form and asymmetric in the access conditions that make use of the form possible. Among the four constraints listed, this is the one most likely to determine whether the primitive's contribution to the population in question is real or merely technical.

Bond-currency jurisdictional embedding. A constraint adjacent to the four above and more easily missed: the bond's denomination currency is itself a jurisdictional artifact. The primitive is denomination-agnostic, but the denominations available in practice are not. The principal US-dollar-pegged stablecoin (USDC) is issued by Circle, a US-jurisdictional entity subject to OFAC sanctions enforcement and capable of freezing addresses on Treasury request; the *Tornado Cash* sanctions of August 2022, which targeted a piece of immutable open-source code rather than an entity, made explicit how far the regulatory reach can extend [OFAC, 2022, Van Valkenburgh, 2022]. ETH is a neutral substrate at the protocol level but is acquired through exchanges that KYC, and on-ramps from local fiat to ETH require banking relationships that the displaced participant typically lacks. A wallet that the bonded primitive cannot reach because the currency-layer gatekeepers refuse to bridge to it is not made reachable by anything the substrate does.

The choice of denomination therefore matters for the displaced-participant case in a way it does not for the counterparty case. We recommend, where the architecture admits it, denominations whose issuance and transfer are not subject to discretionary freeze authority (which excludes most fiat-pegged stablecoins as currently constituted) and whose acquisition does not run through KYC-gated exchanges (which excludes most retail on-ramp paths). In practice the operational candidate is permissionless on-chain assets with deep secondary-market liquidity and no issuer-level freeze authority; the universe is narrower than the universe of all crypto-assets, and the selection criterion is structural rather than ideological.

The cumulative effect of these constraints is that the primitive is not a solved-problem deliverable for displaced populations today. Its present-tense contribution is architectural: commerce-without-recognition is now possible to build, where it was not before, and the constraints that remain are engineering problems with engineering solutions rather than structural features of the legal infrastructure.

Prior art the paper should engage. The constraints listed above are not theoretical: they have been surfaced and partially addressed by humanitarian programs that have piloted blockchain-based assistance over the past decade. The UN High Commissioner for Refugees' *Building Blocks* program, deployed in Jordan since 2017 and expanded across several refugee operations, uses on-chain transaction records to coordinate cash assistance from multiple humanitarian agencies to the same beneficiaries while reducing duplicate disbursement and intermediary cost [UNHCR, 2017, Coppi and Fast, 2019]. The World Food Programme's similar pilots have produced substantial operational experience with the key-custody, fiat-on-ramp, and counterparty-recognition constraints we name above. The independent evaluations [Coppi and Fast, 2019, Caribou Digital, 2018] are mixed: the programs achieve real operational improvements in coordination and disbursement, but the binding constraint identified across evaluations is not architectural recognition of the participant on chain — which the architecture supplies straightforwardly — but the host-state and on-ramp gatekeeping that we list above as distinct constraints. The displaced beneficiaries in these programs do not bond; they receive. The bonded primitive extends the architecture to a use case the existing programs have not yet exercised at scale: the displaced participant as *counterparty* in a productive bilateral exchange rather than as recipient of one-sided disbursement. The empirical record above is informative about what the engineering will require; it is not yet informative about what will happen when displaced participants are bonded counterparties, because the relevant pilots have not been run.

7 Conclusion

The bonded settlement primitive does not solve statelessness, and this paper has not argued that it does. It does change the architecture of the commerce-access gap that statelessness creates, by replacing the precondition of recognized civil-legal subjecthood with the precondition of a cryptographic key. The change is not in the political condition of the populations under discussion; it is in the institutional architecture they face when they attempt to participate in productive activity across jurisdictional boundaries.

The Arendtian distinction is the one to hold: the right to have rights is political, denied by polities, and not within the gift of any technology to restore. The capacity to have commerce is structural, denied by the legal architecture of commerce, and restorable by a settlement primitive whose precondition is cryptographic rather than civil-legal. For populations for whom the first denial is constitutive, the second restoration is non-trivial.

The paper is positional. It identifies a gap, names the architectural change, and acknowledges the operational constraints that remain. The programmatic work — building key-custody infrastructure for refugee populations, designing sponsor pools that lend collateral against verified attestation history, partnering with humanitarian organizations to evaluate where bonded commerce is structurally feasible and where it is not — is for the practitioners who do that work. The contribution of this paper is to make the architectural change legible to scholarship that has not yet recognized it as available.

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